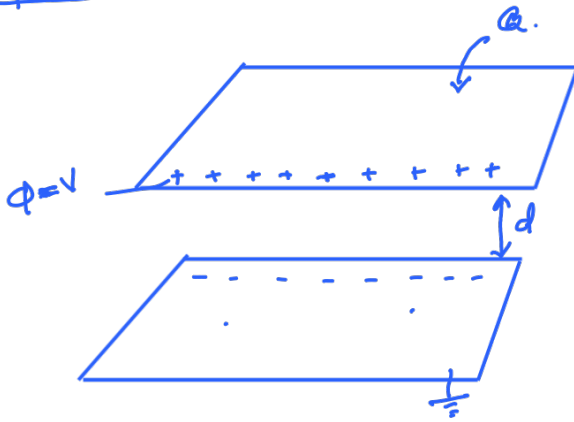


Capacitors:

$$C = \frac{\epsilon_0 A}{d}; \text{ Area of the plates: 'A'}$$

(Fringing field is neglected here.)

$$Q = CV$$

$$\text{Energy stored / work done} = \frac{1}{2} CV^2$$

$$= \frac{1}{2} QV$$

$$\frac{1}{2} \iiint_{\text{vol}} \vec{E} \cdot \vec{D} \, d\tau = \frac{1}{2} \iiint_{\text{vol}} \vec{E} \cdot \vec{\nabla} \phi \, d\tau$$

Since these are conducting plates all the points on the plates are equipotential (ie;  $\phi(\vec{r}) = V$ ) and hence we can say constant potential  $V$  without mentioning position ( $\vec{r}$ ) on the plate.

By poisson's Eqn  $\nabla^2 \phi = -\rho/\epsilon_0$

Assume 1-D case

(Approximation in 1D works as infinite plane)

$$\frac{\partial^2 \phi(x)}{\partial x^2} = -\frac{\rho(x)}{\epsilon_0}$$

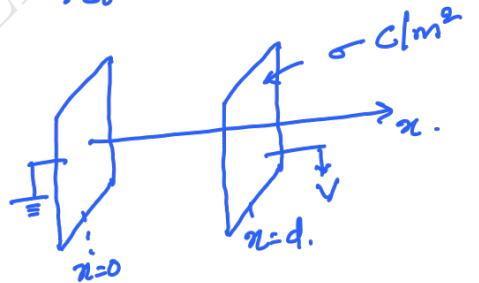
$$\phi(0) = 0; \phi(d) = V$$

$$\text{Also } \frac{\partial^2 \phi}{\partial x^2} = 0 \quad 0 < x < d$$

$$\Rightarrow \phi = A + Bx$$

$$; x=0 \quad \phi(0) = 0 \Rightarrow A = 0$$

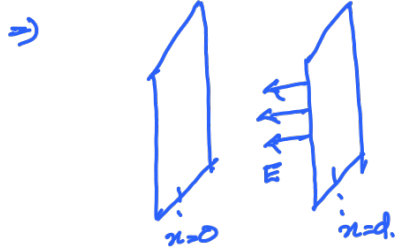
$$x=d; \phi(d) = V \Rightarrow V = Bd \Rightarrow B = V/d$$



when size of capacitor plates  $\gg d$ .  
infinite plane.  
approx ✓

$\therefore \phi = \frac{V}{d} x$ ;  $0 < x < d$  ← This is unique solution.

Now  $\vec{E} = -\nabla\phi \Rightarrow E_x = -\frac{\partial\phi}{\partial x} = -\frac{V}{d}$  (Constant).

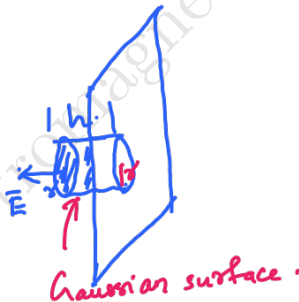


What is the charge on the plates?

Gauss's law;

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q_{enc}}{\epsilon_0}$$

$\vec{E}$  tangential to curved surface of Gaussian cylinder.



Also the other flat surface is inside material (conductor)  $\vec{E} = 0$ . (At least single monolayer thick metal sheets)

So only  $d\vec{s} = \pi r^2 \hat{x}$  contributes to flux.

$$\Rightarrow \left(-\frac{V}{d}\right) \hat{x} \cdot \pi r^2 \hat{x} = \frac{\sigma(\pi r^2)}{\epsilon_0} \leftarrow Q_{enc}$$

$$\Rightarrow \frac{V}{d} \pi r^2 = \frac{\sigma(\pi r^2)}{\epsilon_0} \quad \left( \text{Also we can write } \boxed{|\vec{E}| = \frac{\sigma}{\epsilon_0}} \right)$$

$$\Rightarrow \sigma = \epsilon_0 \frac{V}{d}$$

Example:

$$\text{If } \phi = \frac{V^2}{d^2} x^2.$$

Still it will satisfy boundary condition;

$$\text{but } \frac{d^2\phi}{dx^2} \neq 0.$$

$\Rightarrow$  will not satisfy Laplace's Eqn.

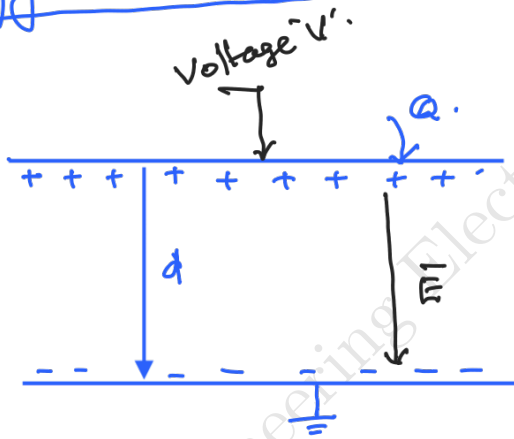
$$Q = \sigma A$$

$$\Rightarrow Q = \frac{\epsilon_0 A}{d} V \quad \text{Capacitance}$$

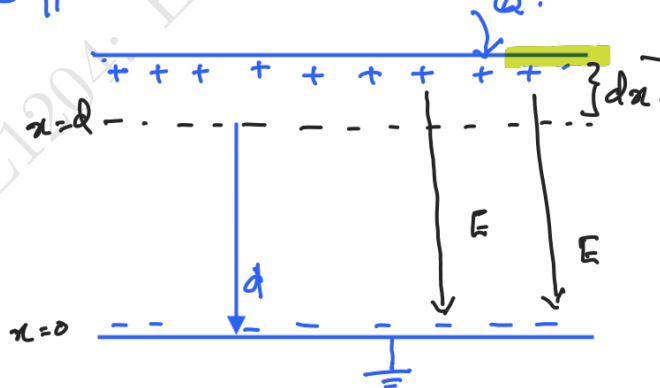
$$= CV \quad ; \quad C = \frac{\epsilon_0 A}{d}$$

In electrostatics, we treat capacitor as an energy storing element.

Energy stored in Capacitor.

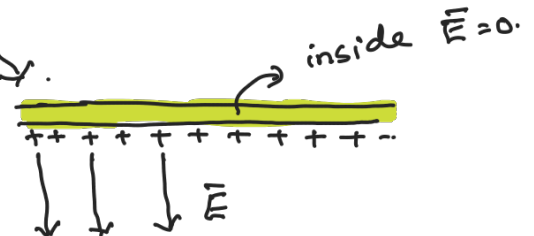


Suppose we move the upper plate to  $d + dx$ .



$$\text{Workdone} = \vec{F} \cdot d\vec{x}$$

$$\vec{F} = ?$$



There for average  $\bar{E}$  in the plate is what need to be used for <sup>calculating</sup> force to move the plate.

$$\Rightarrow \bar{F} = \frac{1}{2}(\bar{E} + 0)Q = \frac{1}{2} \bar{E} Q$$

$$\begin{aligned} \therefore \text{Workdone: Energy stored} &= \bar{F} \cdot d\bar{x} \\ &= \frac{1}{2} \bar{E} Q \cdot d\bar{x} \\ &= \frac{1}{2} E d\bar{x} Q. \\ &= \frac{1}{2} (dV) Q \end{aligned}$$

$$\begin{aligned} W &= \frac{1}{2} E \cdot \frac{A \epsilon_0 d\bar{x}}{\epsilon_0} \\ &= \frac{1}{2} \epsilon_0 E (A d\bar{x}) \end{aligned}$$

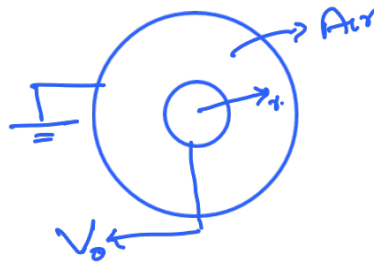
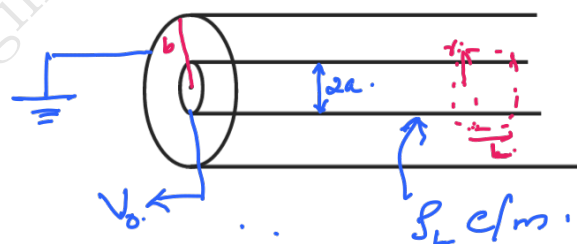
$$\frac{\sigma}{\epsilon_0} = E$$

Volume over which we created  $\bar{E}$  field.

$$\frac{\text{Workdone}}{\text{Volume}} = \frac{1}{2} \epsilon_0 E^2 \rightarrow \text{Energy density (J/m}^3\text{)}$$

$$\boxed{\text{Electrostatic potential Energy } U = \int_{\text{all space}} \frac{1}{2} \epsilon_0 E^2 dv}$$

Capacitance of co-axial cable:



Poisson's Eqn.

$$\nabla^2 \phi = \frac{\rho}{\epsilon_0}$$

$\phi \rightarrow \phi(r)$  ; As no potential variation along 'z' nor along  $\phi$  direction.

$$\Rightarrow \nabla^2 \phi(r) = \frac{\rho}{\epsilon_0}$$

$$\nabla^2 \phi(r) = 0 \quad \text{For } a < r < b.$$

$$\text{Also; } \left. \begin{array}{l} \phi(b) = 0 \\ \phi(a) = V_0 \end{array} \right\} \text{Boundary Condition.}$$

In cylindrical-coordinate

$$\nabla^2 \phi = \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial \phi}{\partial r} \right) + 0 + 0 \leftarrow \frac{\partial \phi}{\partial \theta} = 0, \frac{\partial \phi}{\partial z} = 0.$$

$$\nabla^2 \phi = 0 \Rightarrow \frac{1}{r} \frac{\partial}{\partial r} \left( r \frac{\partial \phi}{\partial r} \right) = 0, \quad r \neq 0.$$

$$\Rightarrow r \frac{d\phi}{dr} = k$$

$$\frac{d\phi}{dr} = \frac{k}{r}$$

$$\phi = k \ln r + A \leftarrow \text{Constant of integration.}$$

$$\phi(b) = 0 \Rightarrow k \ln b + A = 0$$

$$\Rightarrow A = -k \ln b$$

$$\phi(a) = V_0 \Rightarrow k \ln a - k \ln b = V_0.$$

$$\Rightarrow k = \frac{V_0}{\ln(a/b)}$$

$$\therefore \phi = k \ln(r/b) = \frac{V_0 \ln(r/b)}{\ln(a/b)}$$

Find  $\vec{E}$ ?

$$\vec{E}(r) = -\nabla \phi$$

$\nabla\phi$  in cylindrical co-ordinate: (only along  $\hat{r}$ )

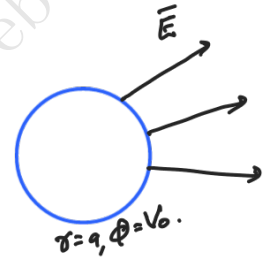
$$\vec{E}(r) = -\frac{\partial\phi}{\partial r} \hat{r}$$

$$= -\frac{\partial}{\partial r} \left[ V_0 \frac{\ln(r/b)}{\ln(a/b)} \right]$$

$$= -\frac{V_0}{\ln(a/b)} \frac{\partial}{\partial r} \ln(r/b) \hat{r}$$

$$= -\frac{V_0}{\ln(a/b)} \frac{1}{r} \hat{r}$$

$$= \frac{-V_0}{r \ln(a/b)} \hat{r} \quad \text{V/m}$$



Overall charge?

$$\oint_S \vec{E} \cdot d\vec{S} = \frac{Q_{enc}}{\epsilon_0}$$

$$\Rightarrow E_r (2\pi r b) = \frac{Q_{enc}}{\epsilon_0}$$

$$\Rightarrow E_r = \frac{Q_{enc}}{2\pi r \epsilon_0}$$

$$\frac{V_0 (2\pi r L)}{r \ln(b/a)} = \frac{Q_{enc}}{\epsilon_0}$$

$$Q = V_0 \frac{2\pi \epsilon_0 L}{\ln(b/a)} \quad \text{Capacitance}$$

$$C = \frac{2\pi \epsilon_0 L}{\ln(b/a)}$$

Usually we write

$$C/L = \frac{2\pi \epsilon_0}{\ln(b/a)}$$

Capacitance/unit length of a coaxial cable  $\Rightarrow$









